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Department of Physics

What Quantum Trace Can a Heavy Particle Leave?

Curvature–Spectator Entanglement
and the Cosmological Collider

Step 5: Gaussian Entanglement

Group-B

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1 Covariance, entropy and the reliability window

For one real sector, $V = \mathrm{Re}(\mathcal{M}\mathcal{M}^\dagger)$ and V_q is its light-field 2×2 block. The reduced symplectic eigenvalue is $\nu_q = \sqrt{\det V_q}$ and $S_q = s(\nu_q)$. The two full-state symplectic eigenvalues must remain $1/2$ because quadratic evolution preserves the purity of the initial state. Outside the horizon, large correlated entries can make the determinant ill conditioned. We first apply the local canonical shear $\widetilde{\Pi}_q = \Pi_q + Q/x$, which leaves $\det V_q$ invariant. Covariance products use extended-precision arithmetic; a local canonical scaling balances the matrix before evaluating its full symplectic eigenvalues. We accept samples with commutator residual $< 10^{-6}$ and full-state eigenvalue error $< 10^{-5}$. The history is shown only up to the first failure of these criteria. For the benchmark, the last sample in this contiguous reliable interval is $x = 5.492 \times 10^{-4}$. The evolution is not extrapolated past it.

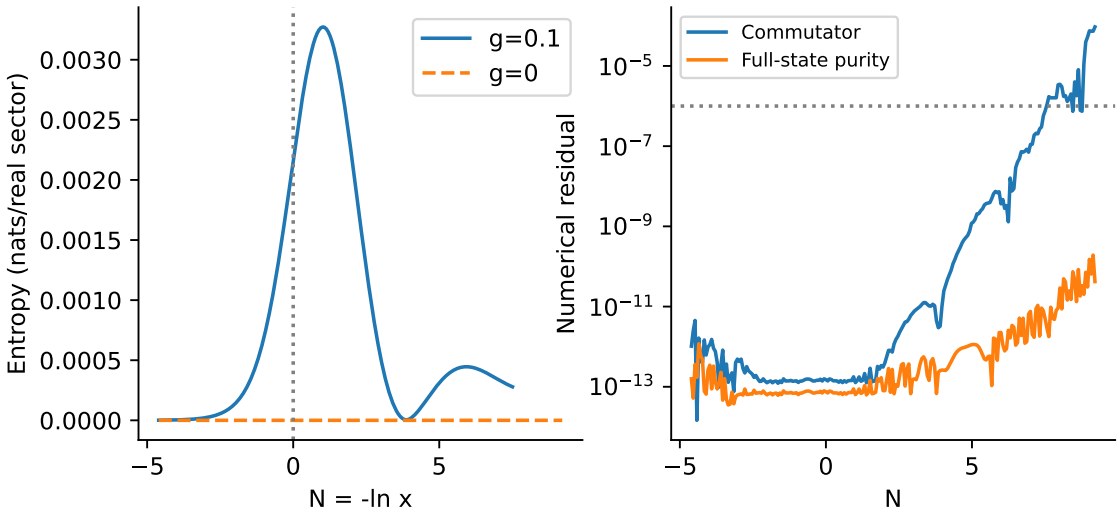


Figure 1: Benchmark entropy and numerical diagnostics. Horizon exit is $N = 0$. The zero-mixing curve measures the numerical floor; it is not physical inter-field entropy.

2 Mass/mixing scan and physical interpretation

We compare $M = 1.6, 1.8, 2.0, 2.4$ and $g = 0, 0.05, 0.1, 0.2$ at the same $x = 1, 0.1, 0.01, 0.001$. All 64 requested comparison rows pass the stated diagnostics. Entropy is quoted per real sector; a generic nonzero momentum pair has twice this value. At $M = 2$, $g = 0.1$, $x = 0.01$, we obtain $S_q = 1.846 \times 10^{-4}$ nats and a fractional light-power increase of 0.003714. Reversing g changes the entropy by 6.003×10^{-12}. This checks the $\sigma \mapsto -\sigma$ symmetry. The largest zero-mixing entropy in the sampled history is 2.227×10^{-9}, which sets a numerical floor rather than an entanglement detection.

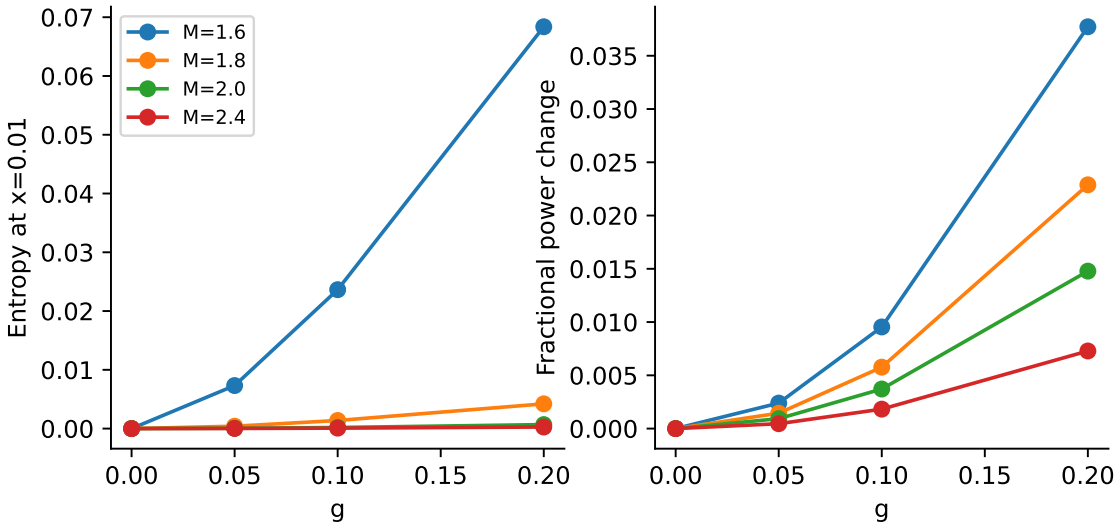


Figure 2: Entropy and fractional power change at the common time $x = 0.01$. They measure different aspects of the Gaussian state.

The time history is not assumed monotone: exchange of correlations can produce growth and decrease while the full state stays pure. These Gaussian entropies omit cubic-interaction corrections and will be compared with a leading cubic bispectrum.

References

[1] C. Weedbrook et al., *Gaussian Quantum Information*, Rev. Mod. Phys. 84, 621 (2012), arXiv:1110.3234.

Code and supporting data: step5 on GitHub.

Contributions (equal)

Member	Contribution
Priyansh Rawat	Covariance and entropy derivation; interpretation.
Samarth Bhatnagar	Purity, zero-mixing and sign-symmetry validation.
Mayank Sinha	Mass/mixing scan, reliability mask and figures.